# Chapter 2

## Image Processing Basics

# What will we learn?

- How is a digital image represented and stored in memory?
- What are the main types of digital image representation?
- What are the most popular image file formats?
- What are the most common types of image processing operations and how do they affect pixel values?

# Digital Image Representation

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0, N-1) \\ f(1,0) & f(1,1) & \cdots & f(1, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1, 0) & f(M-1, 1) & \cdots & f(M-1, N-1) \end{bmatrix}$$

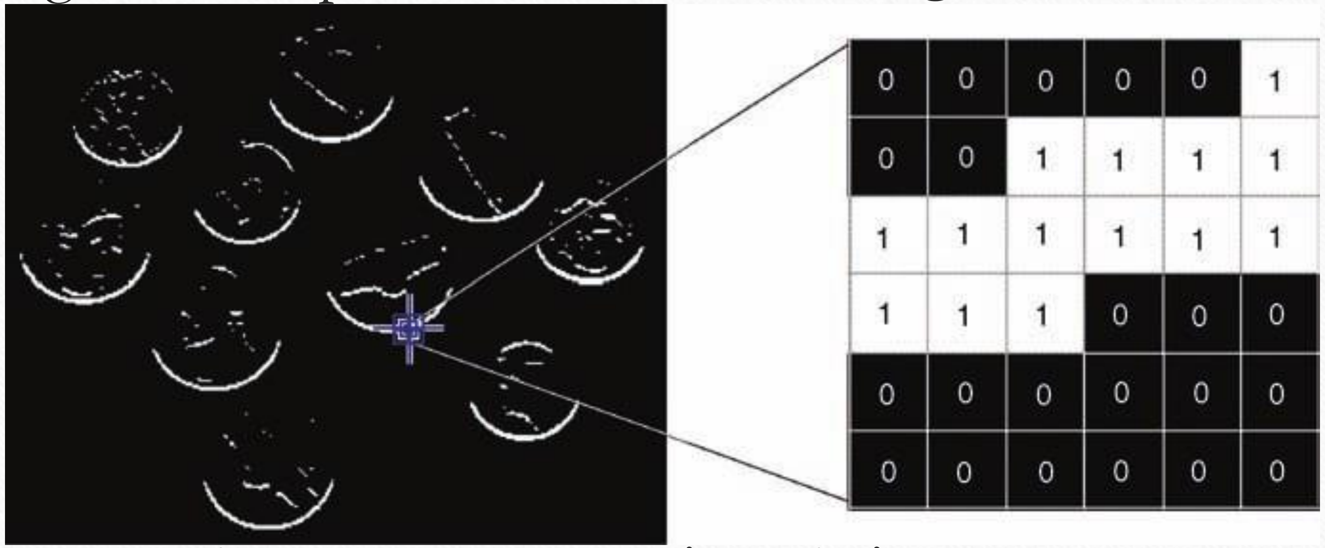
$$\mathtt{f(p,q)} = \begin{bmatrix} \mathtt{f(1,1)} & \mathtt{f(1,2)} & \cdots & \mathtt{f(1,N)} \\ \mathtt{f(2,1)} & \mathtt{f(2,2)} & \cdots & \mathtt{f(2,N)} \\ \vdots & \vdots & & \vdots \\ \mathtt{f(M,1)} & \mathtt{f(M,2)} & \cdots & \mathtt{f(M,N)} \end{bmatrix}$$

# Raster vs Vector

- Raster (or bitmap) image format = one or more 2D arrays of pixels
  - Good quality and fast display speed but larger memory storage requirement and size dependency
- Vector image format = series of drawing commands
  - Resizing without introducing artifact, less size but need to be rasterized for display divices

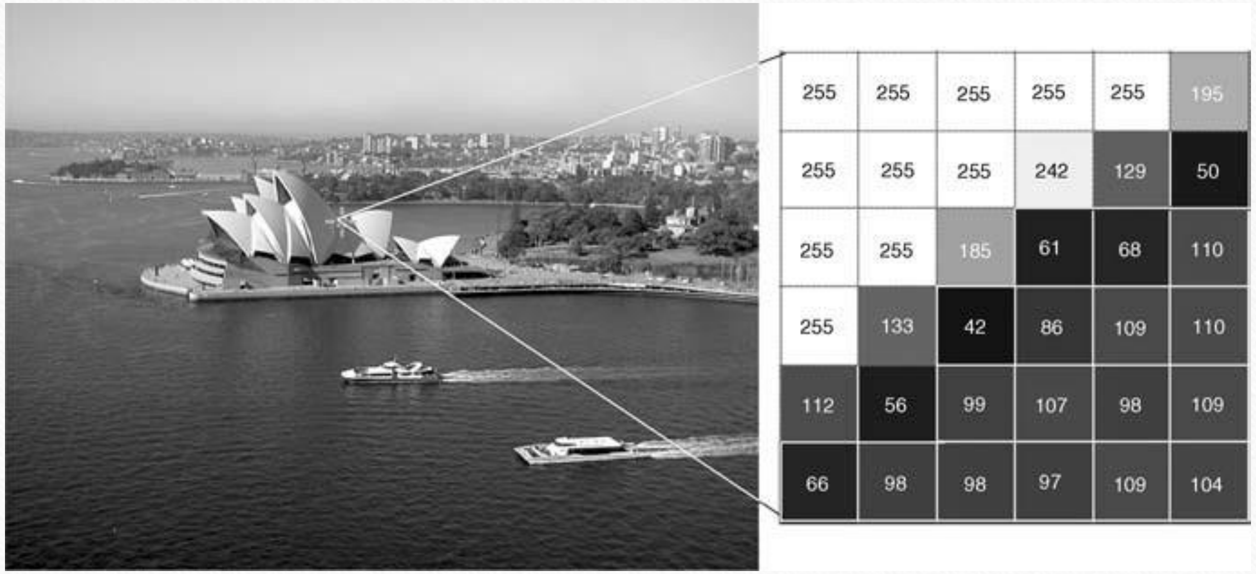
# Example

- A binary image and the pixel values in a 6×6 neighborhood




- In Matlab it is called ***logical*** array of 0's and 1's

# A gray-level image and the pixel values in a 6×6 neighborhood



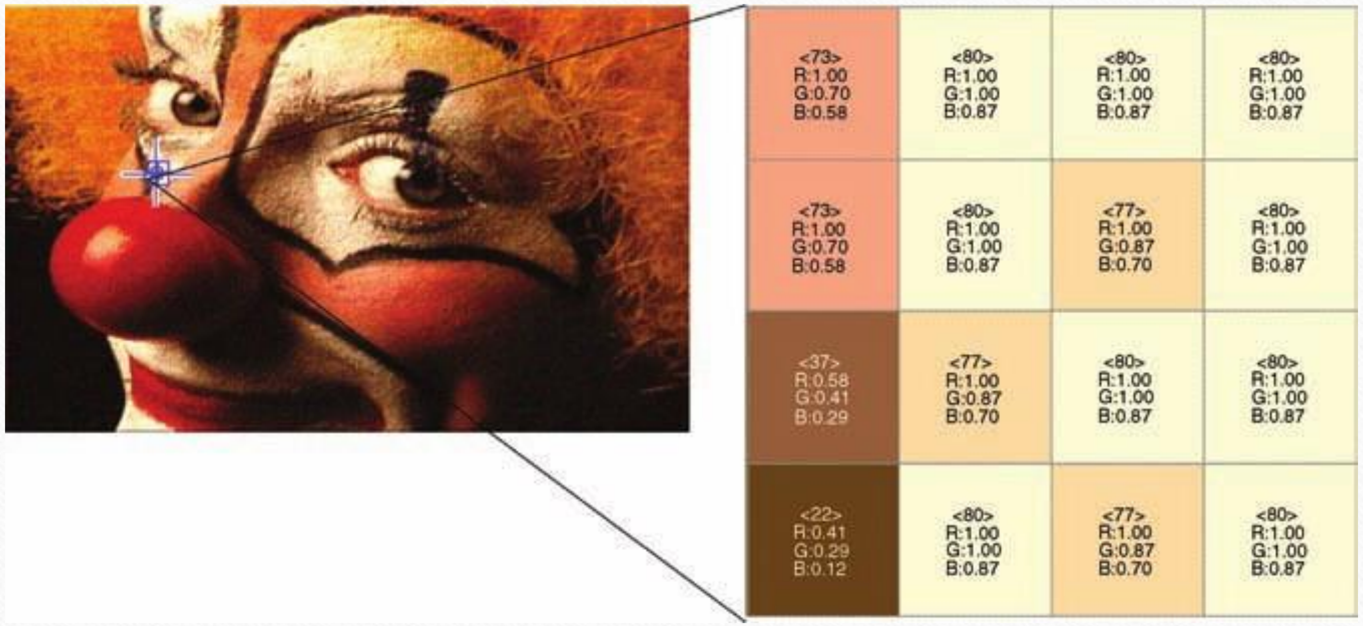

Gray-level, monochrome, grayscale In Matlab most popular are **uint8** ( 0=black, 255=white), **uint16** (0=black, 65535=white), **double** (0.0=black, 1.0=white)

# Color Images

- RGB, a 24-bit number 8bits for each of RED, GREEN and BLUE
- Indexed representation. A pixel contains an index to the pallete of colors…

# Color Image, R, G and B

# Indexed Image...



# Some Image File Formats

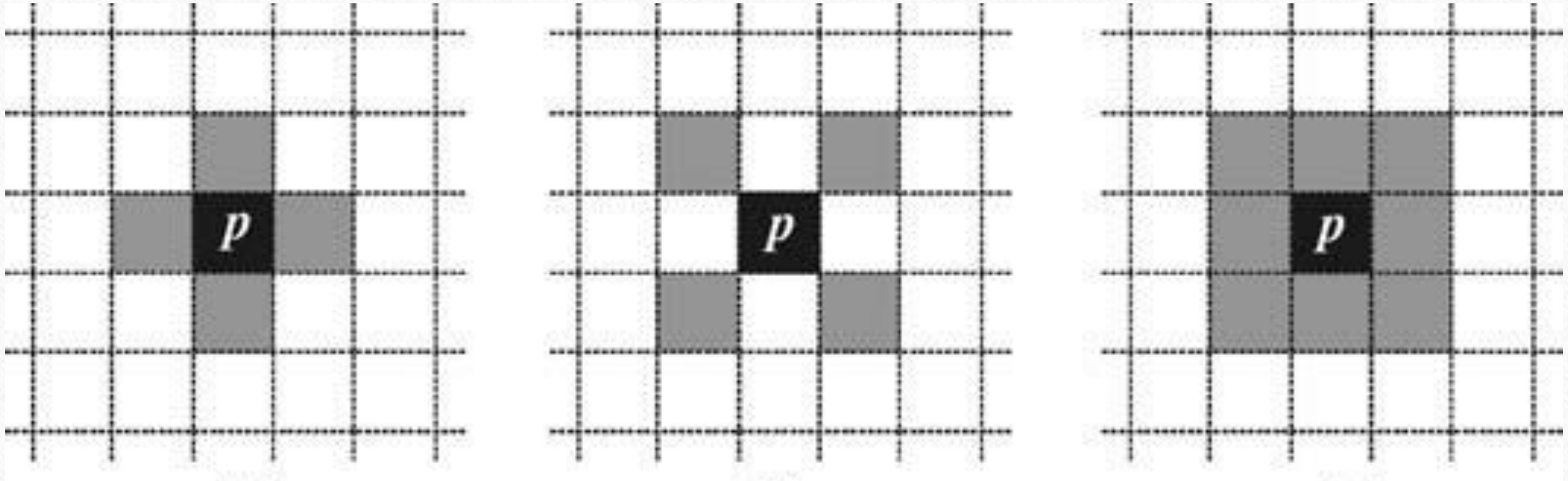
- Lossy vs Lossless Compression
  - BIN, PPM (PBM,PGM, PNM) – raw pixel data

    Need to know all sizes, bytes per pixel etc
  - BMP – Microsoft Windows
  - JPEG – most popular for photographic quality images
  - GIF – uses indexed representation of colors with maximum of 256 colors palette
  - TIFF – 24bit per pixel (truecolor), 5 different compression schemes
  - PNG – supports both indexed and truecolor format, patent free replacement for GIF
  - RAW – adopted by camera manufacturers

# Basic Terminology

- **Image Topology**. It involves the investigation of fundamental image properties, such as number of a particular object, number of separate (not connected) regions, and number of holes in an object etc.
- **Neighborhood**. The pixels surrounding a given pixel constitute its neighborhood, which can be interpreted as a smaller matrix containing (and usually centered around) the reference pixel.

# (a) 4-neighborhood, (b) diagonal neighborhood, (c) 8-neighborhood



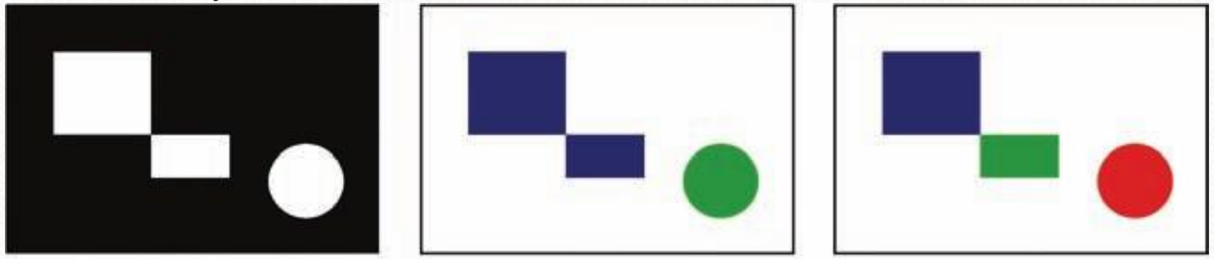

(a) (b) (c)

# Basic Terminology (cont.)

- **Adjacency**: 2 pixel p and q are 4-adjacent if they are 4-neighbors of each other
- 8-adjacency – similar.
- **Paths**. A 4-path between two pixels p and q is a sequence of pixels starting with p and ending with q such that each pixel in the sequence is 4-adjacent to its predecessor in the sequence.
- 8-path – similar.

# Basic Terminology (cont.)

- **Connectivity. If there is a 4-path between pixels p and q,** they are said to be 4-connected
- 8-connected
- **Components. A set of pixels that are connected to each other** is called a component. (a) original (binary) image (b) results for 8-connectivity (c) results for 4-connectivity.



(a) (b) (c)

# Basic Terminology (cont.)

- **Distances Between Pixels**
  - Euclidean distance:

$$D_e(p, q) = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2}$$

  - $D_4$ (also known as *Manhattan* or *city block*) distance:

$$D_4(p, q) = |x_1 - x_0| + |y_1 - y_0|$$

  - $D_8$ (also known as *chessboard*) distance:

$$D_8(p, q) = \max(|x_1 - x_0|, |y_1 - y_0|)$$

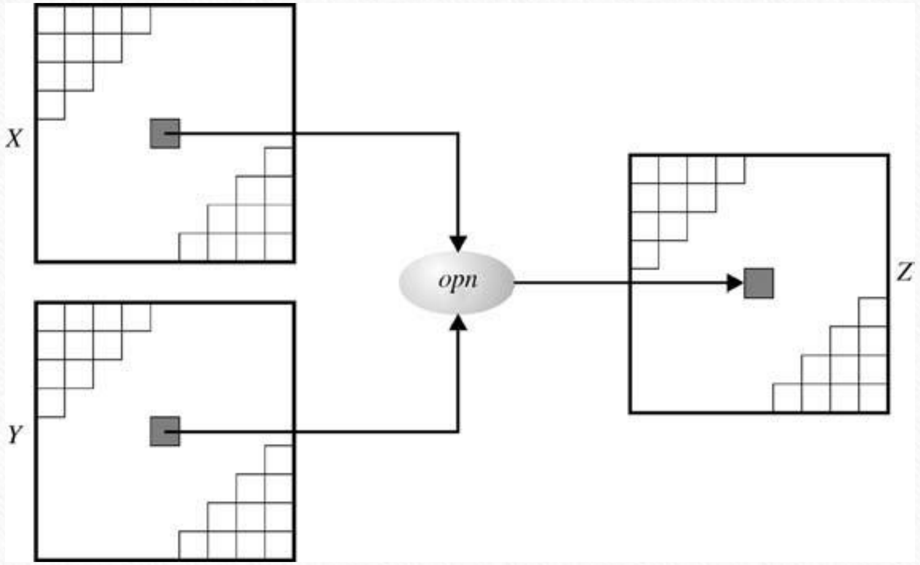
# Image Processing Operations

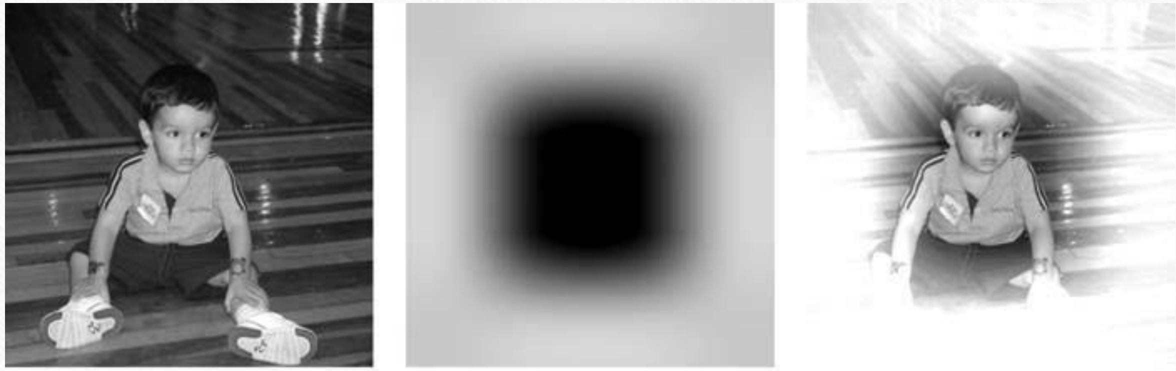
- Operations in the Spatial Domain
  - Global (Point) Operations
  - Neighborhood-Oriented Operations
  - Operations Combining Multiple Images
- Operations in a Transform Domain
  - Fourier Transform
  - Discrete Cosine Transform

# Intensity reduction by 2 (global operation)

(a) (b)

# Combining Two Images

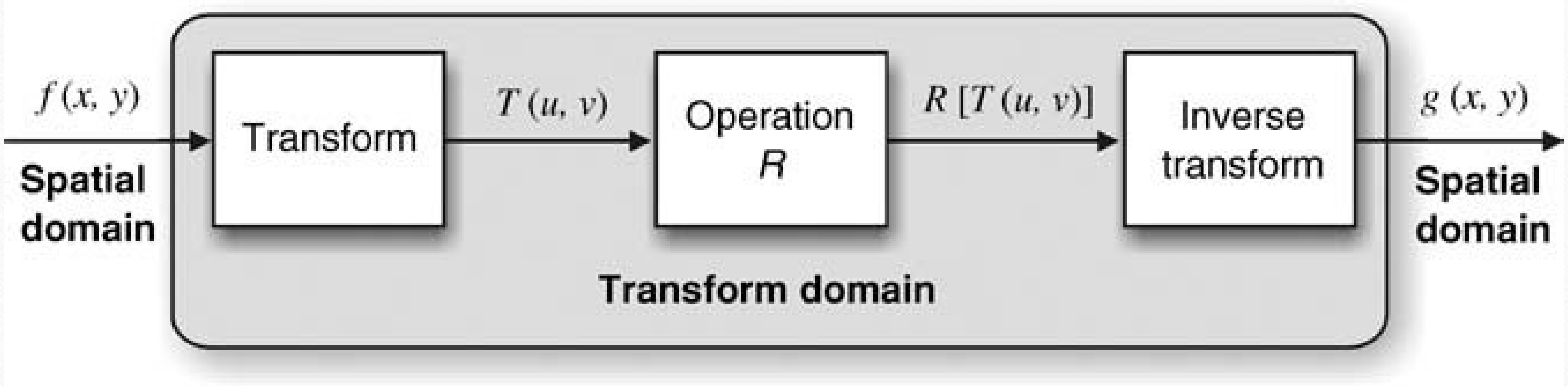


(a) (b) (c)

# Sharpening an image by neighborhood operation

# Operations in Transform Domain



# References


- Practical Image and Video Processing using Matlab, Oge Marques, 2011, **Chapter 2**